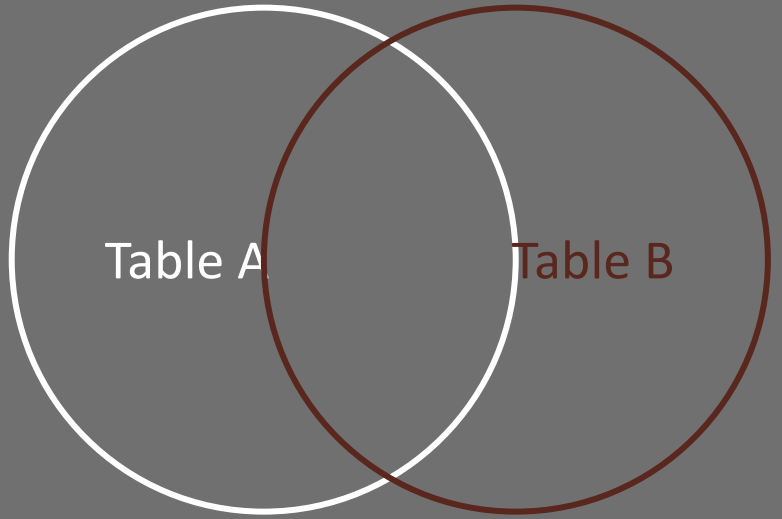
Visual Summary of Joins



Examples adapted from Coding Horror.

A

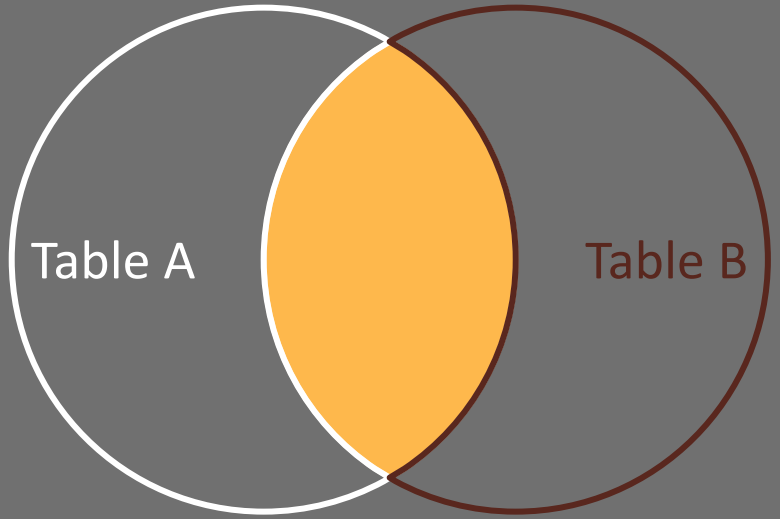
id	data
1	A
1	B
2	C
3	D
4	E
4	F

B

id	data
1	U
2	V
2	W
5	X
5	Y
6	Z

id	A.data	B.data
1	A	U
1	B	U
2	C	V
2	C	W

```
SELECT A.id, A.data, B.data
FROM A
INNER JOIN B ON A.id = B.id;
```



Data must be in both A and B.

A

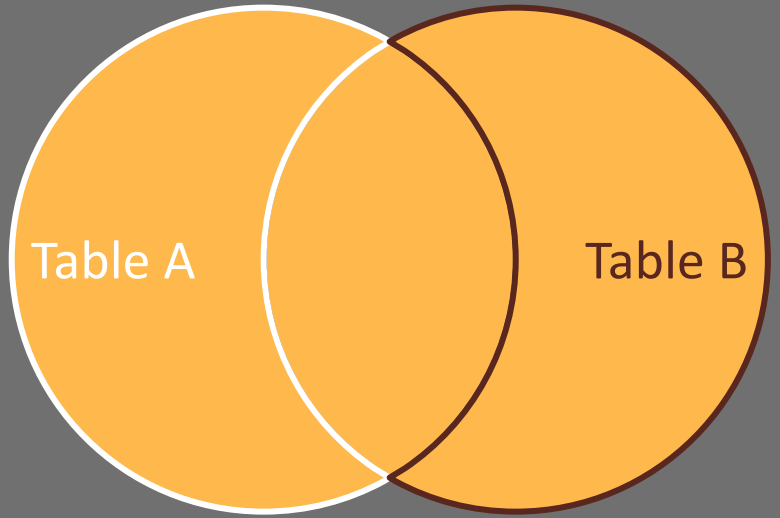
id	data
1	A
1	B
2	C
3	D
4	E
4	F

B

id	data
1	U
2	V
2	W
5	X
5	Y
6	Z

id	A.data	B.data
1	A	U
1	B	U
2	C	V
2	C	W
3	D	NULL
4	E	NULL
4	F	NULL
5	NULL	X
5	NULL	Y
6	NULL	Z

```
SELECT A.id, A.data, B.data
FROM A
FULL OUTER JOIN B ON A.id = B.id;
```



Data in A or B.

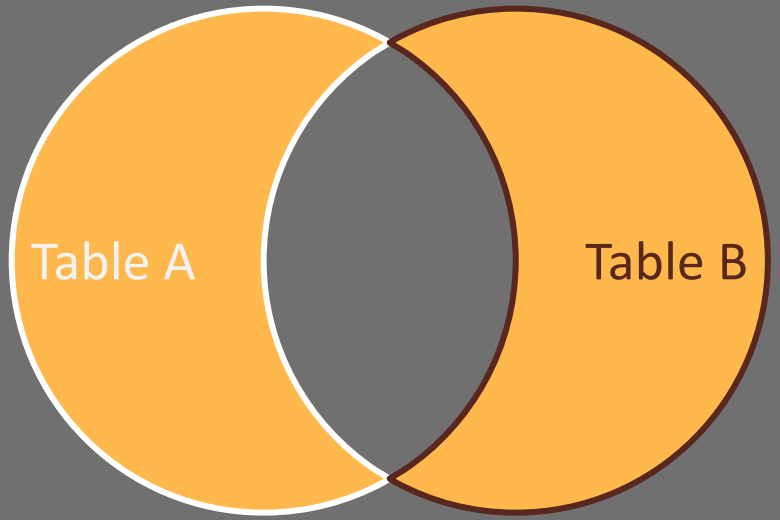
A

id	data
1	A
1	B
2	C
3	D
4	E
4	F

B

id	data
1	U
2	V
2	W
5	X
5	Y
6	Z

id	A.data	B.data
3	D	NULL
4	E	NULL
4	F	NULL
5	NULL	X
5	NULL	Y
6	NULL	Z




```
SELECT A.id, A.data, B.data
FROM A
FULL OUTER JOIN B ON A.id = B.id
WHERE A.data IS NULL OR B.data IS NULL;
```

Data in either A or B, but not both.

A

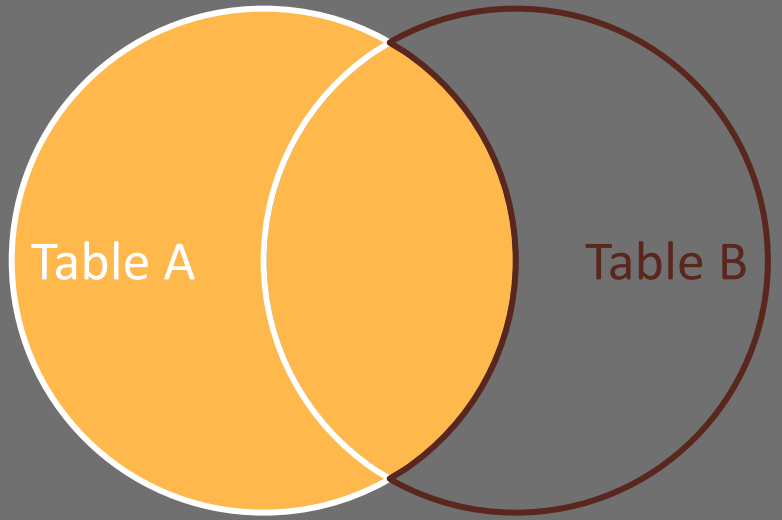
id	data
1	A
1	B
2	C
3	D
4	E
4	F

B

id	data
1	U
2	V
2	W
5	X
5	Y
6	Z

id	A.data	B.data
1	A	U
1	B	U
2	C	V
2	C	W
3	D	NULL
4	E	NULL
4	F	NULL

```
SELECT A.id, A.data, B.data
FROM A
LEFT OUTER JOIN B ON A.id = B.id;
```



Data in A.

A

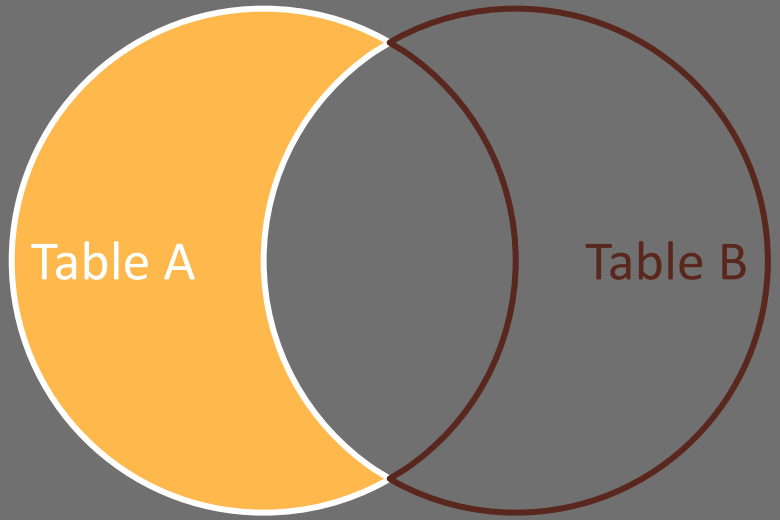
id	data
1	A
1	B
2	C
3	D
4	E
4	F

B

id	data
1	U
2	V
2	W
5	X
5	Y
6	Z

id	A.data	B.data
3	D	NULL
4	E	NULL
4	F	NULL

```
SELECT A.id, A.data, B.data
FROM A
LEFT OUTER JOIN B ON A.id = B.id
WHERE B.data IS NULL;
```



Data in only A.

Final notes about joins

- Avoid implicit join conditions or keywords.

- Here: different join operations
- Later: different join implementations